The Gauges of Potential and Field

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Gauge and Potential Formula

1. **Gauge Transformations**:

$$\begin{pmatrix} \mathbf{A} \\ V \end{pmatrix} \mapsto \begin{cases} \mathbf{A}' = \mathbf{A} + \boldsymbol{\nabla}\lambda \\[2ex] V' = V - \dfrac{\partial \lambda}{\partial t} \end{cases}$$

2. **Different Gauge**:

$$\begin{cases} \text{Coulomb Gauge} : \boldsymbol{\nabla} \cdot \mathbf{A} = 0 \\[2ex] \text{Lorenz Gauge} : \boldsymbol{\nabla} \cdot \mathbf{A} = -\dfrac{1}{c^2}\dfrac{\partial V}{\partial t} \end{cases}$$

3. **d'Alembertian**:

$$\Box^2 \equiv \nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}$$

4. **Retarded and Advanced time**

$$\text{Retarded Time} : t_r = t - \frac{\mathscr{r}}{c}, \quad \text{Advantaged Time} : t_a = t + \frac{\mathscr{r}}{c}$$

5. **Point Charge**:$\rho(\mathbf{r}, t_r) = q\,\delta^{(3)}\big(\mathbf{r} - \mathbf{w}(t_r)\big) = q\,\delta^{(3)}(\boldsymbol{\mathscr{r}})$

$$\begin{cases} V(\mathbf{r}, t) = \dfrac{1}{4\pi\epsilon_0}\dfrac{q}{\mathscr{r}(1 - \mathbf{v}/c \cdot \hat{\boldsymbol{\mathscr{r}}})} \\[2ex] \mathbf{A}(\mathbf{r}, t) = \dfrac{\mu_0}{4\pi}\dfrac{\mathbf{I}(t_r)}{\mathscr{r}(1 - \mathbf{v}/c \cdot \hat{\boldsymbol{\mathscr{r}}}} = \dfrac{\mathbf{v}}{c^2}V(\mathbf{r}, t) \end{cases}$$

$$\begin{cases} \mathbf{E}(\mathbf{r}, t) = \dfrac{q}{4\pi\epsilon_0}\dfrac{\mathscr{r}}{(\boldsymbol{\mathscr{r}} \cdot \mathbf{u})^3}\Big[(c^2 - v^2)\mathbf{u} + \underbrace{\boldsymbol{\mathscr{r}} \times (\mathbf{u} \times \mathbf{a})}_{\text{Radiation}}\Big] \\[2ex] \mathbf{B}(\mathbf{r}, t) = \dfrac{1}{c}\hat{\boldsymbol{\mathscr{r}}} \times \mathbf{E}(\mathbf{r}, t) \end{cases}$$

1 Gauge Symmetry and Gauge Fixing

1.1 Gauge Symmetry: Redundancy of Description

In electromagnetism, the potentials $(V, \mathbf{A})$ are not uniquely determined by the physical fields $(\mathbf{E}, \mathbf{B})$. The fundamental transformation is given by

$$V \to V' = V - \partial_t \lambda, \qquad \mathbf{A} \to \mathbf{A}' = \mathbf{A} + \nabla \lambda, \tag{1}$$

where $\lambda = \lambda(\mathbf{r}, t)$ is an arbitrary function.

A direct computation shows that

$$\mathbf{E}' = \mathbf{E}, \qquad \mathbf{B}' = \mathbf{B}. \tag{2}$$

$$\boxed{\text{Gauge transformation does not change } (\mathbf{E}, \mathbf{B}).}$$

This implies that two sets of potentials related by a gauge transformation correspond to the same physical configuration. Therefore,

$$\boxed{\text{Gauge symmetry is not a physical symmetry, but a redundancy of description.}}$$

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1.2 Gauge vs Global Symmetry

Consider instead a spacetime translation

$$\mathbf{r} \to \mathbf{r} + \mathbf{b}, \qquad t \to t + a. \tag{3}$$

Such a transformation changes measurable quantities such as positions and distances. Hence,

$$\boxed{\text{Global symmetry transforms one physical state into another.}}$$

The essential distinction is therefore not the form of transformation, but its physical effect:

$$\boxed{\text{A transformation is a gauge symmetry if and only if all physical observables remain unchanged.}}$$

1.3 Local Nature of Gauge Symmetry

Gauge transformations are parametrized by a function $\lambda(\mathbf{r}, t)$, which carries infinitely many degrees of freedom. In contrast, global symmetries involve only a finite number of parameters.

$$\boxed{\text{Gauge symmetry} = \text{infinite-dimensional redundancy.}}$$

However, this distinction is not fundamental. In certain contexts (e.g. general relativity or periodic boundary conditions), even transformations with constant parameters can behave as gauge transformations. The true criterion always lies in physical observables.

1.4 Physical Observables and Their Limitations

A physical observable is defined as a quantity measurable in experiments. Two configurations are physically equivalent if all observables coincide.

A natural question arises:

Does $(\mathbf{E}, \mathbf{B})$ *completely determine the physical state?*

The answer is not always, as demonstrated by the Aharonov–Bohm effect.

1.5 Aharonov–Bohm Effect and Wilson Loop

Consider a region where

$$\mathbf{E} = \mathbf{B} = 0, \tag{4}$$

but $\mathbf{A} \neq 0$. Although classical fields vanish, quantum interference experiments detect a phase shift. This effect is captured by the Wilson loop

$$W(C) = \exp\left(i \oint_C \mathbf{A} \cdot \mathrm{d}\mathbf{r} \right). \tag{5}$$

$\boxed{W(C) \text{ is gauge invariant and physically observable.}}$

If the loop C bounds a surface S, then

$$\oint_C \mathbf{A} \cdot \mathrm{d}\mathbf{r} = \int_S \mathbf{B} \cdot \mathrm{d}\mathbf{a}. \tag{6}$$

However, in a space with nontrivial topology, there exist loops that cannot be shrunk to a point. In such cases,

$\boxed{(\mathbf{E}, \mathbf{B}) \text{ are insufficient to characterize the physical state.}}$

Instead, global information encoded in $\mathbf{A}$ becomes physically relevant.

1.6 Gauge Fixing: Eliminating Redundancy

Since multiple potentials correspond to the same physical fields, solving Maxwell's equations in terms of $(V, \mathbf{A})$ leads to non-uniqueness.

$\boxed{\text{Gauge fixing selects a unique representative from each equivalence class.}}$

Mathematically, this corresponds to imposing a constraint on $(V, \mathbf{A})$.

1.7 Criteria for a Good Gauge Fixing Condition

A gauge fixing condition must satisfy two fundamental requirements.

Existence. For any physical configuration $(\mathbf{E}, \mathbf{B})$, there must exist a function λ such that the transformed potentials satisfy the condition.

No physical state should be excluded.

Uniqueness. The condition should eliminate as much redundancy as possible. Ideally, it establishes a one-to-one correspondence between potentials and fields.

Gauge fixing should remove all non-physical degrees of freedom.

1.8 Operational Verification

To verify a gauge fixing condition, one must perform:

1. **Existence test:** Solve for λ such that the transformed potentials satisfy the condition.
2. **Uniqueness test:** Show that the only λ preserving the condition is $\lambda = 0$ (up to boundary conditions).

1.9 Examples and Non-Examples

Valid gauge conditions.

- Coulomb gauge:
$$\nabla \cdot \mathbf{A} = 0 \tag{7}$$
- Lorenz gauge:
$$\nabla \cdot \mathbf{A} + \frac{1}{c^2}\partial_t V = 0 \tag{8}$$

These conditions can always be reached by solving differential equations for λ.

Too strong conditions.

- $\mathbf{A} = 0$
- $\partial_x A_y - \partial_y A_x = 0$

The second condition is equivalent to
$$\partial_x A_y - \partial_y A_x = B_z. \tag{9}$$

Gauge transformations cannot change $\mathbf{B}$.

Hence, such conditions eliminate physical configurations and violate the existence requirement.

Too weak conditions.

- $\int \mathrm{d}t\, V = 0$

  This removes only a single parameter while leaving infinitely many gauge freedoms.

1.10 Conceptual Summary

Gauge symmetry represents redundancy.
Gauge fixing removes redundancy without altering physics.
Physical observables determine equivalence classes, not the potentials themselves.

2 Gauge

2.1 The Potential Fixing

In electrostatics, the electric field $\mathbf{E}$ can be represented as

$$\mathbf{E} = -\boldsymbol{\nabla} V \quad \boldsymbol{\nabla} \times \mathbf{E} = \boldsymbol{\nabla} \times (-\boldsymbol{\nabla} V) = \mathbf{0},$$

but in electrodynamics, it's not only $-\boldsymbol{\nabla} V$ anymore. Since $\boldsymbol{\nabla} \cdot \mathbf{B} = 0$, $\exists \mathbf{A}$ such that $\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}$. By Maxwell's Equations

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \qquad \boldsymbol{\nabla} \cdot \mathbf{B} = 0$$
$$\boldsymbol{\nabla} \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

the curl of $\mathbf{E}$ becomes

$$\boldsymbol{\nabla} \times \mathbf{E} = -\frac{\partial}{\partial t} \boldsymbol{\nabla} \times \mathbf{A} \Rightarrow \boldsymbol{\nabla} \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0 \tag{10}$$

Since the curl vanishes, we can rewrite the term into

$$\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} = -\boldsymbol{\nabla} V$$

which gives that

$$\mathbf{E} = -\boldsymbol{\nabla} V - \frac{\partial \mathbf{A}}{\partial t}. \tag{11}$$

Therefore, Gauss's law shows that the divergence of the field is

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \boldsymbol{\nabla} \cdot \left(-\boldsymbol{\nabla} V - \frac{\partial \mathbf{A}}{\partial t} \right) = -\nabla^2 V - \frac{\partial}{\partial t} \boldsymbol{\nabla} \cdot \mathbf{A} = \frac{\rho}{\epsilon_0},$$

and we obtain

$$\nabla^2 V + \frac{\partial}{\partial t}\boldsymbol{\nabla}\cdot\mathbf{A} = -\frac{\rho}{\epsilon_0}. \tag{12}$$

Last but not least, the Amphère's law is rewritten in terms of $\mathbf{A}$ and V:

$$\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\mathbf{A}) = \boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\mathbf{A}) - \nabla^2\mathbf{A} = \mu_0\mathbf{J} + \mu_0\epsilon_0\frac{\partial}{\partial t}\left(-\boldsymbol{\nabla}V - \frac{\partial\mathbf{A}}{\partial t}\right),$$

The rearrangement leads to

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\mathbf{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) - \left(\nabla^2\mathbf{A} - \mu_0\epsilon_0\frac{\partial^2\mathbf{A}}{\partial t^2}\right) = \mu_0\mathbf{J}. \tag{13}$$

So, the modified electric field arising from the changing of magnetic field do construct the good model to contain all the information in Maxwell's equations. So, the key point is that: *How can I choose* $\mathbf{A}$*? How many fields can satisfy these equations and properties?*

2.2 Potential

To verify (2), first we suppose that we have to different fields pair $(\mathbf{A}, V)$ and $(\mathbf{A}', V')$ which correspond to the same field $\mathbf{E}, \mathbf{B}$.
We let

$$\mathbf{A}' = \mathbf{A} + \boldsymbol{\alpha}, \quad V' = V + \beta. \tag{14}$$

Since they correspond to the same fields, so they must satisfy

$$\begin{cases} \boldsymbol{\nabla}\times\mathbf{A}' = \boldsymbol{\nabla}\times(\mathbf{A}+\boldsymbol{\alpha}) \Rightarrow \boldsymbol{\nabla}\times\boldsymbol{\alpha} = \mathbf{0}, \\ -\boldsymbol{\nabla}V' - \dfrac{\partial\mathbf{A}'}{\partial t} = -\boldsymbol{\nabla}(V+\beta) - \dfrac{\partial\mathbf{A}}{\partial t} - \dfrac{\partial\boldsymbol{\alpha}}{\partial t} \Rightarrow \boldsymbol{\nabla}\beta + \dfrac{\partial\boldsymbol{\alpha}}{\partial t} = 0. \end{cases} \tag{15}$$

By these relations, we can find a function λ to satisfy

$$\boldsymbol{\alpha} = \boldsymbol{\nabla}\lambda \tag{16}$$

since the curl of $\boldsymbol{\alpha}$ is 0. Substitute $\boldsymbol{\nabla}\lambda$ into the second term, we have

$$\boldsymbol{\nabla}\left(\beta + \frac{\partial\lambda}{\partial t}\right) = 0, \tag{17}$$

which leads to

$$\beta = -\frac{\partial\lambda}{\partial t}. \tag{18}$$

In general, there must be a function $\Lambda(t)$ which satisfies:

$$\beta = -\frac{\partial\lambda'}{\partial t} + \Lambda(t), \tag{19}$$

which leads to

$$\lambda = \lambda' + \int_0^{t'} \Lambda(t')\,\mathrm{d}t. \tag{20}$$

So, the choices of the function become more, or say, infinitely many since

$$\lambda \mapsto \beta \mapsto \alpha. \tag{21}$$

Therefore, the quantities have **Gauge Freedom**, with different gauge, the correspond **Observables** will not be changed. Substitute (18) into (14), we have:

$$\begin{cases} \mathbf{A}' = \mathbf{A} + \boldsymbol{\nabla}\lambda \\ V' = V - \dfrac{\partial \lambda}{\partial t} \end{cases} \longleftrightarrow \text{Same } \mathbf{E} \text{ \& } \mathbf{B} \tag{22}$$

So, the changes of the potentials which won't affect the physical quantities **E** and **B** are called **Gauge Transformation**. The following will show two useful gauge fixing conditions.

2.3 Coulomb Gauge

The Coulomb gauge is:

$$\boxed{\boldsymbol{\nabla} \cdot \mathbf{A} = 0.} \tag{23}$$

This gauge fixing condition will reduce some degree of freedom to make the problem easier to solve (but **A** is hard to solve). By taking this gauge, (12) becomes the Poisson's equation, and (13) will also change:

$$\begin{cases} \nabla^2 V = -\dfrac{\rho}{\epsilon_0}, \\ \nabla^2 \mathbf{A} - \mu_0\epsilon_0 \dfrac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu_0 \mathbf{J} + \mu_0\epsilon_0 \boldsymbol{\nabla}\left(\dfrac{\partial V}{\partial t}\right). \end{cases} \tag{24}$$

Note that in Poisson's equation, it only consider the spacial variables such that the potential is

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t_{\text{now}})}{\mathscr{r}}\,\mathrm{d}\tau',$$

which is determined by the distribution of charges change Right Now. That is, the distribution instantaneous determine the field in the whole space. It violates the relativity (it's impossible that it can instantaneously do such thing since it needs speed faster than light).

2.4 Lorenz Gauge

The Lorenz gauge is given by

$$\boxed{\boldsymbol{\nabla} \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}.} \tag{25}$$

This gauge makes the equations (12) and (13) become

$$\begin{cases} \nabla^2 \mathbf{A} - \mu_0 \epsilon_0 \dfrac{\partial^2}{\partial t^2} \mathbf{A} = -\mu_0 \mathbf{J} \\ \nabla^2 V - \mu_0 \epsilon_0 \dfrac{\partial^2}{\partial t^2} V = -\dfrac{\rho}{\epsilon_0}, \end{cases} \tag{26}$$

and we can define the same differential operator, which is called **d'Alembertian**

$$\boxed{\Box^2 \equiv \nabla^2 - \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2}.} \tag{27}$$

Therefore, (26) can be written in terms of $\Box^2$:

$$\boxed{\begin{cases} \Box^2 \mathbf{A} = -\mu_0 \mathbf{J} \\ \Box^2 V = -\dfrac{\rho}{\epsilon_0}. \end{cases}} \tag{28}$$

In general, the Lorenz gauge is defined by the condition

$$\partial_\mu A^\mu = 0$$

which treats space and time symmetrically. Unlike the Coulomb gauge, which separates space and time and leads to a Poisson equation for the scalar potential, the Lorenz gauge leads to

$$\Box^2 A^\mu = \mu_0 J^\mu$$

so that both ϕ and $\mathbf{A}$ satisfy inhomogeneous wave equations. This means the potentials are not determined by the instantaneous distribution of sources, but instead evolve dynamically in spacetime.

More importantly, the solutions naturally take the form of retarded potentials,

$$t_r = t - \frac{|\mathbf{r} - \mathbf{r}'|}{c}$$

so that the potentials at $(\mathbf{r}, t)$ depend only on the source at earlier times. In this sense, the Lorenz gauge is fully consistent with relativistic causality, and does not involve any instantaneous influence. Rather than describing only the present configuration like the Coulomb gauge, it provides a spacetime-consistent description where information propagates at finite speed c.

3 Continuous Distribution

3.1 Concept of Retarded

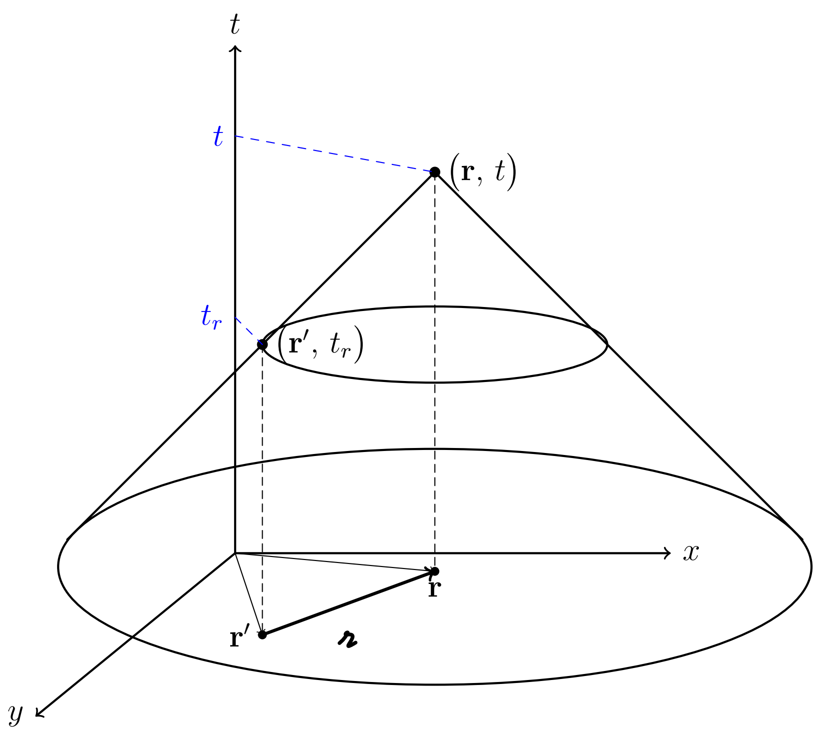
Since the Lorenz gauge implies the **nonstatic** charges, the potential is not instantaneous anymore. The potential is now

$$\boxed{\begin{cases} V(\mathbf{r},t) = \dfrac{1}{4\pi\epsilon_0}\displaystyle\int \mathrm{d}\tau' \, \frac{\rho(\mathbf{r}', t_r)}{\mathscr{r}} \\ \mathbf{A}(\mathbf{r},t) = \dfrac{\mu_0}{4\pi}\displaystyle\int \mathrm{d}\tau' \, \frac{\mathbf{J}(\mathbf{r}', t_r)}{\mathscr{r}} \end{cases}} \tag{29}$$

for continuous distribution. The variable t_r is called Retarded time. Since the distance between the position of charges $\mathbf{r}'$ affecting the observation point($\mathbf{r}$) from $\mathbf{r}$ is $\mathscr{r}$, so the light must travel this distance, it must **delay**. So, what affects the point now $(\mathbf{r}, t)$ is the charges some time ago. Just like the sun we see is its appearance eight minutes ago. The time light need to travel is $\mathscr{r}/c$, so the retarded time is

$$\boxed{t_r \equiv t - \frac{\mathscr{r}}{c}.} \tag{30}$$

And the charges at the retarded time is call **Retarded charges**. We can visualize the retarded charge in 3D space with z axis becomes time. As shown below:



This figure illustrates the distribution of charges that influence the observation point at retarded times. Imagine a circle centered around the observer in a plane: charges located farther away require a longer time for light to propagate, so the corresponding retarded time t_r becomes smaller. As we consider circles of increasing radius, these combine to form a conical structure in spacetime. The charges that contribute to the field at the observation point therefore lie on the surface of this cone, known as the light cone. In fact, the three-dimensional visualization of the light cone corresponds to circular slices in the xy-plane; in full three-dimensional space, the distribution of retarded charges forms spherical shells.

Furthermore, if we perform the transformation $t \to -t$, the light cone flips upward. In this case, we must use the **advanced time**, defined as

$$\boxed{t_a \equiv t + \frac{\mathscr{r}}{c}} \tag{31}$$

which describes the present configuration as determined by charges in the future. The two light cones exhibit a symmetry; the only difference lies in the reversal of causality in the solutions, much like playing a video in reverse.

3.2 Jefimenko's Equations

Although the solution to the potentials with the retarded time are similar to the potential before. However, the electric/magnetic fields are not

$$\mathbf{E} \neq \frac{1}{4\pi\epsilon_0}\int d\tau' \frac{\rho(\mathbf{r}', t_r)}{\mathscr{r}^2}\hat{\mathscr{r}}, \quad \mathbf{B} \neq \frac{\mu_0}{4\pi}\int d\tau' \frac{\mathbf{J}(\mathbf{r}', t_r) \times \hat{\mathscr{r}}}{\mathscr{r}^2}$$

Note that the charge density depends not only on the position but also time. Take gradient of V, we obtain

$$\nabla V = \frac{1}{4\pi\epsilon_0}\int d\tau' \Big[(\nabla\rho)\frac{1}{\mathscr{r}} + \rho\nabla \underbrace{\left(\frac{1}{\mathscr{r}}\right)}_{-\hat{\mathscr{r}}/\mathscr{r}^2}\Big], \tag{32}$$

where $\nabla\rho$ is[1]

$$\nabla\rho(\mathbf{r}', t_r) = \dot{\rho}\nabla t_r = \dot{\rho}\nabla\left(t - \frac{\mathscr{r}}{c}\right) = -\frac{1}{c}\dot{\rho}\underbrace{\nabla\mathscr{r}}_{=\hat{\mathscr{r}}}. \tag{33}$$

Hence, (32) becomes:

$$\nabla V = \frac{1}{4\pi\epsilon_0}\int d\tau' \left[-\frac{\dot{\rho}}{c}\frac{\hat{\mathscr{r}}}{\mathscr{r}} - \rho\frac{\hat{\mathscr{r}}}{\mathscr{r}^2}\right]. \tag{34}$$

[1]Note that this result is different from Liénard–Wiechert case since we consider $\mathscr{r} = \mathbf{r} - \mathbf{r}'$ and the term $\mathbf{r}'$ is an integral variables.

By (11), and the definition of vector potential, we've known that

$$\mathbf{E} = -\boldsymbol{\nabla} V - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A},$$

so we still have one term to deal with:

$$\frac{\partial}{\partial t}\mathbf{A} = \frac{\partial}{\partial t}\frac{\mu_0}{4\pi}\int \mathrm{d}\tau' \, \frac{\mathbf{J}(\mathbf{r}', t_r)}{\mathscr{r}} = \frac{\mu_0}{4\pi}\int \mathrm{d}\tau' \, \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{\mathscr{r}} = \frac{1}{4\pi\epsilon_0}\int \mathrm{d}\tau' \, \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{c^2\mathscr{r}}$$

combine this with (34), we can find the correct electric field is

$$\boxed{\mathbf{E}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0}\int \mathrm{d}\tau' \left[\frac{\rho(\mathbf{r}', t_r)}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}} + \frac{\dot{\rho}(\mathbf{r}', t_r)}{c\mathscr{r}}\hat{\boldsymbol{\mathscr{r}}} - \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{c^2\mathscr{r}}\right]} \tag{35}$$

Similarly, the magnetic field is

$$\boxed{\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi}\int \mathrm{d}\tau' \left[\frac{\mathbf{J}(\mathbf{r}', t_r)}{\mathscr{r}^2} + \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{c\mathscr{r}}\right] \times \hat{\boldsymbol{\mathscr{r}}}} \tag{36}$$

These relation between the retarded potential and fields are called **Jefimenko's Equations**.

4 Point Charges

4.1 Liénard-Wiechert Potentials

We now derive the scalar potential $V(\mathbf{r}, t)$ and vector potential $\mathbf{A}(\mathbf{r}, t)$ produced by a moving point charge. Let the trajectory of the charge be $\mathbf{w}(t)$, and let its velocity be(at the retarded time)

$$\mathbf{v}(t_r) = \dot{\mathbf{w}}(t_r).$$

Note:
The velocity of the charge at the retarded time is the differentiation of $\mathbf{w}(t_r)$ with respect to r_r, that is

$$\frac{\mathrm{d}\mathbf{w}(t_r)}{\mathrm{d}t_r} = \mathbf{v}(t_r).$$

While the time we observe fields is t, and the fields we measured is from the sources at the retarded time t_r. Namely, if the observer change the time t, the position of the retarded charge $\mathbf{w}$ correspondingly change, and so does the retarded time. So, if we differentiate $\mathbf{w}$ with respect to t, we have:

$$\frac{\partial \mathbf{w}(t_r)}{\partial t} = \frac{\mathrm{d}\mathbf{w}(t)}{\mathrm{d}t_r}\frac{\partial t_r}{\partial t} = \mathbf{v}\frac{\partial t_r}{\partial t}. \tag{37}$$

Since $t_r = t - \mathscr{r}/c$, we have:

$$\frac{\partial t_r}{\partial t} = 1 - \frac{1}{c}\frac{\partial \mathscr{r}}{\partial t}, \tag{38}$$

where $\mathscr{r} = |\mathbf{r} - \mathbf{w}(t_r)|$, which also depends on t_r, hence:

$$\frac{\partial \mathscr{r}}{\partial t} = \frac{\partial \mathscr{r}}{\partial t_r}\frac{\partial t_r}{\partial t} = \frac{\partial}{\partial t_r}\left(\boldsymbol{\mathscr{r}} \cdot \boldsymbol{\mathscr{r}}\right)^{1/2}\frac{\partial t_r}{\partial t}, \tag{39}$$

where

$$\begin{aligned}\frac{\partial}{\partial t_r}\left(\boldsymbol{\mathscr{r}} \cdot \boldsymbol{\mathscr{r}}\right)^{1/2} &= \left(\frac{1}{2\mathscr{r}}\right)\frac{\partial \boldsymbol{\mathscr{r}}}{\partial t_r} \cdot \boldsymbol{\mathscr{r}} \times 2, \quad (\boldsymbol{\mathscr{r}} = \mathbf{r}(t) - \mathbf{w}(t_r)) \\ &= \frac{1}{\mathscr{r}}\hat{\boldsymbol{\mathscr{r}}} \cdot \big[- \mathscr{r}\mathbf{v}(t_r)\big] = -\mathbf{v}(t_r) \cdot \hat{\boldsymbol{\mathscr{r}}}.\end{aligned} \tag{40}$$

Substituting (40) into (39), we have:

$$\frac{\partial \mathscr{r}}{\partial t} = \Big[- \mathbf{v}(t_r) \cdot \hat{\boldsymbol{\mathscr{r}}}\Big]\frac{\partial t_r}{\partial t}, \tag{41}$$

again, substituting (41) into (38) and rearranging the equation, we obtain:

$$\frac{\partial t_r}{\partial t}\left[1 - \frac{\mathbf{v}(t_r)}{c} \cdot \hat{\boldsymbol{\mathscr{r}}}\right] = 1 \Rightarrow \frac{\partial t_r}{\partial t} = \frac{1}{\left(1 - \hat{\boldsymbol{\mathscr{r}}} \cdot \mathbf{v}/c\right)}. \tag{42}$$

Back to the equation (37), the velocity of the retarded source(charge) related to observer, on the other hand, with respect to t is:

$$\boxed{\frac{\partial \mathbf{w}(t_r)}{\partial t} = \frac{\mathbf{v}}{\left(1 - \hat{\boldsymbol{\mathscr{r}}} \cdot \mathbf{v}/c\right)}} \tag{43}$$

In Lorenz gauge, the retarded potentials are

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0}\int \mathrm{d}\tau' \int \mathrm{d}t' \, \frac{\delta(t - t' - \mathscr{r}/c)}{\mathscr{r}} \, \rho(\mathbf{r}', t'), \tag{44}$$

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi}\int \mathrm{d}\tau' \int \mathrm{d}t' \, \frac{\delta(t - t' - \mathscr{r}/c)}{\mathscr{r}} \, \mathbf{J}(\mathbf{r}', t'). \tag{45}$$

Here we use

$$\boldsymbol{\mathscr{r}} \equiv \mathbf{r} - \mathbf{r}', \qquad \mathscr{r} \equiv |\mathbf{r} - \mathbf{r}'|, \qquad \hat{\boldsymbol{\mathscr{r}}} \equiv \frac{\boldsymbol{\mathscr{r}}}{\mathscr{r}}. \tag{46}$$

For a point charge q moving along $\mathbf{w}(t)$, the charge density and current density are

$$\rho(\mathbf{r}', t') = q\,\delta^{(3)}\big(\mathbf{r}' - \mathbf{w}(t')\big), \tag{47}$$

$$\mathbf{J}(\mathbf{r}', t') = q\,\mathbf{v}(t')\,\delta^{(3)}\big(\mathbf{r}' - \mathbf{w}(t')\big). \tag{48}$$

We first substitute $\rho(\mathbf{r}', t')$ into the scalar potential:

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \mathrm{d}\tau' \int \mathrm{d}t' \, \frac{\delta(t - t' - \boldsymbol{\mathscr{r}}/r)}{\mathscr{r}} \, q\,\delta^{(3)}\big(\mathbf{r}' - \mathbf{w}(t')\big). \tag{49}$$

The integration over $\mathbf{r}'$ is immediate:

$$V(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \int \mathrm{d}t' \, \frac{\delta(t - t' - |\mathbf{r} - \mathbf{w}(t')|/c)}{|\mathbf{r} - \mathbf{w}(t')|}.$$

Now define

$$\mathbf{R}(t') \equiv \mathbf{r} - \mathbf{w}(t'), \quad R(t') \equiv |\mathbf{R}(t')|, \quad f(t') \equiv t - t' - \frac{R(t')}{c}. \tag{50}$$

Then

$$V(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \int \mathrm{d}t' \, \frac{\delta\big(f(t')\big)}{R(t')}. \tag{51}$$

We now use the one dimensional delta function identity

$$\delta\big(f(t')\big) = \sum_i \frac{\delta(t' - t_i)}{|f'(t_i)|},$$

where t_i are the roots of $f(t') = 0$. For the retarded solution, we choose the retarded root t_r, which satisfies

$$f(t_r) = 0 \Rightarrow t_r = t - \frac{R(t_r)}{c}.$$

At the retarded time, we define

$$\boldsymbol{\mathscr{r}} \equiv \mathbf{r} - \mathbf{w}(t_r), \quad \mathscr{r} \equiv |\boldsymbol{\mathscr{r}}|, \quad \hat{\boldsymbol{\mathscr{r}}} \equiv \frac{\boldsymbol{\mathscr{r}}}{\mathscr{r}}. \tag{52}$$

Next we compute $f'(t')$.

$$f(t') = t - t' - \frac{R(t')}{c} \Rightarrow f'(t') = -1 - \frac{1}{c}\frac{\mathrm{d}R}{\mathrm{d}t'}. \tag{53}$$

Also,

$$\mathbf{R}(t') = \mathbf{r} - \mathbf{w}(t') \Longrightarrow \frac{\mathrm{d}\mathbf{R}}{\mathrm{d}t'} = -\mathbf{v}(t').$$

Using $R = |\mathbf{R}|$ we obtain

$$\frac{\mathrm{d}R}{\mathrm{d}t'} = \frac{\mathbf{R}}{R} \cdot \frac{\mathrm{d}\mathbf{R}}{\mathrm{d}t'} = \hat{\mathbf{R}} \cdot (-\mathbf{v}) = -\hat{\mathbf{R}} \cdot \mathbf{v}. \tag{54}$$

Therefore,

$$f'(t') = -1 + \frac{\hat{\mathbf{R}} \cdot \mathbf{v}}{c}. \tag{55}$$

Evaluating at $t' = t_r$, where $\hat{\mathbf{R}} = \hat{\boldsymbol{\mathscr{r}}}$, gives

$$f'(t_r) = -\left(1 - \frac{\mathbf{v}}{c} \cdot \hat{\boldsymbol{\mathscr{r}}}\right) \Rightarrow |f'(t_r)| = 1 - \frac{\mathbf{v}}{c} \cdot \hat{\boldsymbol{\mathscr{r}}}. \tag{56}$$

Thus the delta function becomes

$$\delta\big(f(t')\big) = \frac{\delta(t' - t_r)}{1 - \mathbf{v}/c \cdot \hat{\boldsymbol{\mathscr{r}}}}.$$

Substituting this into the integral for $V(\mathbf{r}, t)$, we get

$$V(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \int \mathrm{d}t' \, \frac{1}{R(t')} \frac{\delta(t' - t_r)}{1 - \mathbf{v}/c \cdot \hat{\boldsymbol{\mathscr{r}}}}. \tag{57}$$

Now the t' integration is immediate:

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \frac{q}{\mathscr{r} \left(1 - \mathbf{v}/c \cdot \hat{\boldsymbol{\mathscr{r}}}\right)}.$$

This is the Liénard–Wiechert scalar potential:

$$\boxed{V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \frac{q}{\mathscr{r} \left(1 - \frac{\mathbf{v}}{c} \cdot \hat{\boldsymbol{\mathscr{r}}}\right)}}. \tag{58}$$

We now derive the vector potential. Substituting $\mathbf{J}(\mathbf{r}', t')$ into the retarded formula gives

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \mathrm{d}\tau' \int \mathrm{d}t' \, \frac{\delta(t - t' - \mathscr{r}/c)}{\mathscr{r}} \, q \, \mathbf{v}(t') \, \delta^{(3)}\big(\mathbf{r}' - \mathbf{w}(t')\big). \tag{59}$$

Again performing the $\mathbf{r}'$ integration, we obtain

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0 q}{4\pi} \int \mathrm{d}t' \, \frac{\mathbf{v}(t')}{|\mathbf{r} - \mathbf{w}(t')|} \, \delta\left(t - t' - \frac{|\mathbf{r} - \mathbf{w}(t')|}{c}\right).$$

Using the same function $f(t')$ and the same delta function identity as above,

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0 q}{4\pi} \int \mathrm{d}t' \, \frac{\mathbf{v}(t')}{R(t')} \, \delta\big(f(t')\big) = \frac{\mu_0 q}{4\pi} \int \mathrm{d}t' \, \frac{\mathbf{v}(t')}{R(t')} \, \frac{\delta(t' - t_r)}{1 - \mathbf{v}/c \cdot \hat{\boldsymbol{\mathscr{r}}}}.$$

Therefore,

$$\mathbf{A}(\mathbf{r},t) = \frac{\mu_0 q}{4\pi}\frac{\mathbf{v}(t_r)}{\mathscr{r}\left(1-\mathbf{v}/c\cdot\hat{\boldsymbol{\mathscr{r}}}\right)}. \tag{60}$$

If we define

$$\mathbf{I}(t_r) \equiv q\,\mathbf{v}(t_r), \tag{61}$$

then this becomes

$$\boxed{\mathbf{A}(\mathbf{r},t) = \frac{\mu_0}{4\pi}\frac{\mathbf{I}(t_r)}{\mathscr{r}\left(1-\frac{\mathbf{v}}{c}\cdot\hat{\boldsymbol{\mathscr{r}}}\right)}}. \tag{62}$$

We may rewrite $\mathbf{A}$ as

$$\mathbf{A}(\mathbf{r},t) = \frac{\mu_0 q}{4\pi}\frac{\mathbf{v}}{\mathscr{r}\left(1-\mathbf{v}/c\cdot\hat{\boldsymbol{\mathscr{r}}}\right)} = \frac{1}{4\pi\epsilon_0}\frac{q\,\mathbf{v}}{c^2\,\mathscr{r}\left(1-\mathbf{v}/c\cdot\hat{\boldsymbol{\mathscr{r}}}\right)} = \frac{\mathbf{v}}{c^2}V(\mathbf{r},t).$$

Hence

$$\boxed{\mathbf{A}(\mathbf{r},t) = \frac{\mathbf{v}}{c^2}V(\mathbf{r},t)}. \tag{63}$$

Here every quantity $\mathbf{v}$, $\boldsymbol{\mathscr{r}}$, $\mathscr{r}$, and $\hat{\boldsymbol{\mathscr{r}}}$ is understood to be evaluated at the retarded time t_r, determined implicitly by

$$t_r = t - \frac{\mathscr{r}}{c}. \tag{64}$$

The double integration over space and time reflects a fundamental physical fact: electromagnetic influence propagates at a finite speed.

In electrostatics, the potential depends only on the instantaneous charge distribution, so we integrate over space at a fixed time. However, in electrodynamics, the field at $(\mathbf{r},t)$ cannot depend on the source at the same time t, but only on information that has had enough time to travel to $\mathbf{r}$.

Therefore, we must consider all possible emission events $(\mathbf{r}',t')$ in spacetime. The integral

$$\int \mathrm{d}\tau' \int \mathrm{d}t'$$

represents a scan over all source points and all emission times. Each such event contributes a signal that propagates at speed c. The delta function enforces causality by selecting only those events that lie on the past light cone. Indeed, the condition

$$t - t' = \frac{|\mathbf{r}-\mathbf{r}'|}{c}$$

means that only signals emitted at time t' from position $\mathbf{r}'$, which can reach $\mathbf{r}$ exactly at time t, are allowed to contribute. Thus, the retarded time is not assumed, but emerges

naturally from the delta function constraint.
After performing the time integral, the solution collapses to a single contribution at

$$t' = t_r, \qquad t_r = t - \frac{\mathscr{r}}{c},$$

and only at this stage do we define

$$\boldsymbol{\mathscr{r}} = \mathbf{r} - \mathbf{w}(t_r).$$

In summary, the Liénard–Wiechert potentials encode the idea that fields are determined by the history of the source, not its instantaneous state.

4.2 Fields

The Liénard–Wiechert potentials can be denoted by

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \frac{q}{\mathscr{r}(1 - \hat{\boldsymbol{\mathscr{r}}} \cdot \mathbf{v}/c)}, \quad \mathbf{A}(\mathbf{r}, t) = \frac{\mathbf{v}}{c^2} V(\mathbf{r}, t).$$

And the corresponding fields of dynamic system are

$$\mathbf{E} = -\boldsymbol{\nabla} V - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}.$$

Part 1. We first evaluate the term $\boldsymbol{\nabla} V$:
Take the gradient of the potential V, we can obtain

$$\boldsymbol{\nabla} V = \frac{1}{4\pi\epsilon_0} \frac{-q}{\mathscr{r}^2(1 - \hat{\boldsymbol{\mathscr{r}}} \cdot \mathbf{v}/c)^2} \boldsymbol{\nabla}\Big[\mathscr{r}(1 - \hat{\boldsymbol{\mathscr{r}}} \cdot \mathbf{v}/c)\Big]. \tag{65}$$

The blue part can be written as

$$\boldsymbol{\nabla}\Big[\mathscr{r}(1 - \hat{\boldsymbol{\mathscr{r}}} \cdot \mathbf{v}/c)\Big] = \boldsymbol{\nabla}\mathscr{r} - \boldsymbol{\nabla}(\boldsymbol{\mathscr{r}} \cdot \mathbf{v}/c) = \boldsymbol{\nabla}\mathscr{r} - \frac{1}{c}\boldsymbol{\nabla}(\boldsymbol{\mathscr{r}} \cdot \mathbf{v})$$

since $t_r = t - \mathscr{r}/c$, then

$$\boldsymbol{\nabla} t_r = -\frac{1}{c}\boldsymbol{\nabla}\mathscr{r} \Rightarrow \boldsymbol{\nabla}\mathscr{r} = -c\boldsymbol{\nabla} t_r. \tag{66}$$

This can be rewritten as

$$\begin{aligned} \boldsymbol{\nabla}\mathscr{r} &= \frac{1}{2\mathscr{r}}\boldsymbol{\nabla}(\boldsymbol{\mathscr{r}} \cdot \boldsymbol{\mathscr{r}}) = \frac{1}{\mathscr{r}}\Big[\boldsymbol{\mathscr{r}} \times (\boldsymbol{\nabla} \times \boldsymbol{\mathscr{r}}) + (\boldsymbol{\mathscr{r}} \cdot \boldsymbol{\nabla})\boldsymbol{\mathscr{r}}\Big] \\ &= \frac{1}{\mathscr{r}}\Big[\boldsymbol{\mathscr{r}} \times (\boldsymbol{\nabla} \times \mathbf{r}) - \boldsymbol{\mathscr{r}} \times (\boldsymbol{\nabla} \times \mathbf{w}) + (\boldsymbol{\mathscr{r}} \cdot \boldsymbol{\nabla})\mathbf{r} - (\boldsymbol{\mathscr{r}} \cdot \boldsymbol{\nabla})\mathbf{w}\Big] \end{aligned} \tag{67}$$

We need to write down everything in terms of $\boldsymbol{\nabla} t_r$. The expression of the other term by product rule is:

$$\frac{1}{c}\boldsymbol{\nabla}(\boldsymbol{\mathscr{r}} \cdot \mathbf{v}) = \frac{1}{c}\Big[\boldsymbol{\mathscr{r}} \times (\boldsymbol{\nabla} \times \mathbf{v}) + \mathbf{v} \times (\boldsymbol{\nabla} \times \boldsymbol{\mathscr{r}}) + (\boldsymbol{\mathscr{r}} \cdot \boldsymbol{\nabla})\mathbf{v} + (\mathbf{v} \cdot \boldsymbol{\nabla})\boldsymbol{\mathscr{r}}\Big]. \tag{68}$$

(a) $\boldsymbol{\mathscr{r}} \times (\boldsymbol{\nabla} \times \mathbf{v})$: Deal with $\boldsymbol{\nabla} \times \mathbf{v}$ first, and since the velocity here is $\mathbf{v}(t_r)$, we need to apply chain rule:

$$\begin{aligned}
\boldsymbol{\nabla} \times \mathbf{v} &= \hat{\mathbf{x}} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) + \hat{\mathbf{y}} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) + \hat{\mathbf{z}} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \\
&= \hat{\mathbf{x}} \left(\frac{\mathrm{d}v_z}{\mathrm{d}t_r} \frac{\partial t_r}{\partial y} - \frac{\mathrm{d}v_y}{\mathrm{d}t_r} \frac{\partial t_r}{\partial z} \right) + \hat{\mathbf{y}} \left(\frac{\mathrm{d}v_x}{\mathrm{d}t_r} \frac{\partial t_r}{\partial z} - \frac{\mathrm{d}v_z}{\mathrm{d}t_r} \frac{\partial t_r}{\partial x} \right) + \hat{\mathbf{z}} \left(\frac{\mathrm{d}v_y}{\mathrm{d}t_r} \frac{\partial t_r}{\partial x} - \frac{\mathrm{d}v_x}{\mathrm{d}t_r} \frac{\partial t_r}{\partial y} \right) \\
&= -\mathbf{a}(t_r) \times \boldsymbol{\nabla} t_r
\end{aligned}$$

where we define $\mathbf{a} = \mathrm{d}\mathbf{v}/\mathrm{d}t_r$. Therefore,

$$\boldsymbol{\mathscr{r}} \times (\boldsymbol{\nabla} \times \mathbf{v}) = \boldsymbol{\mathscr{r}} \times (-\mathbf{a} \times \boldsymbol{\nabla} t_r). \tag{69}$$

(b) $\mathbf{v} \times (\boldsymbol{\nabla} \times \boldsymbol{\mathscr{r}})$: Since $\boldsymbol{\mathscr{r}} = \mathbf{r} - \mathbf{w}$, the curl $\boldsymbol{\nabla} \times \boldsymbol{\mathscr{r}} = \boldsymbol{\nabla} \times \mathbf{r} - \boldsymbol{\nabla} \times \mathbf{w} = -\boldsymbol{\nabla} \times \mathbf{w}$. Just like what I did in (69), we can obtain the left curl:

$$\mathbf{v} \times (\boldsymbol{\nabla} \times \boldsymbol{\mathscr{r}}) = \mathbf{v} \times \Big(- \underbrace{\boldsymbol{\nabla} \times \mathbf{w}}_{-\mathbf{v} \times \boldsymbol{\nabla} t_r} \Big) = \mathbf{v} \times (\mathbf{v} \times \boldsymbol{\nabla} t_r). \tag{70}$$

(c) $(\boldsymbol{\mathscr{r}} \cdot \boldsymbol{\nabla})\mathbf{v}$: Expand the dot product:

$$(\boldsymbol{\mathscr{r}} \cdot \boldsymbol{\nabla})\mathbf{v} = \left(\mathscr{r}_x \frac{\partial}{\partial x} + \mathscr{r}_y \frac{\partial}{\partial y} + \mathscr{r}_z \frac{\partial}{\partial z} \right) \mathbf{v}(t_r),$$

by chain rule,

$$\begin{aligned}
(\boldsymbol{\mathscr{r}} \cdot \boldsymbol{\nabla})\mathbf{v} &= \left(\mathscr{r}_x \frac{\partial \mathbf{v}}{\partial x} + \mathscr{r}_y \frac{\partial \mathbf{v}}{\partial y} + \mathscr{r}_z \frac{\partial \mathbf{v}}{\partial z} \right) = \left(\mathscr{r}_x \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t_r} \frac{\partial t_r}{\partial x} + \mathscr{r}_y \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t_r} \frac{\partial t_r}{\partial y} + \mathscr{r}_z \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t_r} \frac{\partial t_r}{\partial z} \right) \\
&= \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t_r} \left(\mathscr{r}_x \frac{\partial t_r}{\partial x} + rc_y \frac{\partial t_r}{\partial y} + \mathscr{r}_z \frac{\partial t_r}{\partial z} \right) = \mathbf{a}(\boldsymbol{\mathscr{r}} \cdot \boldsymbol{\nabla} t_r).
\end{aligned} \tag{71}$$

(d) $(\mathbf{v} \cdot \boldsymbol{\nabla})\boldsymbol{\mathscr{r}}$: Similar to (b), we can separate the term $\boldsymbol{\mathscr{r}}$ into $\mathbf{r} - \mathbf{w}$. Therefore

$$(\mathbf{v} \cdot \boldsymbol{\nabla})\boldsymbol{\mathscr{r}} = (\mathbf{v} \cdot \boldsymbol{\nabla})\mathbf{r} - \underbrace{(\mathbf{v} \cdot \boldsymbol{\nabla})\mathbf{w}}_{\text{Same logic as (c)}} = \underbrace{\left(v_x \frac{\partial \mathbf{r}}{\partial x} + v_y \frac{\partial \mathbf{r}}{\partial y} + v_z \frac{\partial \mathbf{r}}{\partial z} \right)}_{\hat{\mathbf{x}} v_x + \hat{\mathbf{y}} v_y + \hat{\mathbf{z}} v_z} - \mathbf{v}(\mathbf{v} \cdot \boldsymbol{\nabla} t_r).$$

Finally, we obtain:

$$(\mathbf{v} \cdot \boldsymbol{\nabla})\boldsymbol{\mathscr{r}} = \mathbf{v} - \mathbf{v}(\mathbf{v} \cdot \boldsymbol{\nabla} t_r) \tag{72}$$

Substituting these four results into (68), we obtain:

$$\begin{aligned}
\frac{1}{c}\boldsymbol{\nabla}(\boldsymbol{\mathscr{r}}\cdot\mathbf{v}) &= \frac{1}{c}\Big[-\boldsymbol{\mathscr{r}}\times(\mathbf{a}\times\underline{\boldsymbol{\nabla}t_r})+\mathbf{v}\times(\mathbf{v}\times\underline{\boldsymbol{\nabla}t_r})+\mathbf{a}(\boldsymbol{\mathscr{r}}\cdot\underline{\boldsymbol{\nabla}t_r})+\mathbf{v}-\mathbf{v}(\mathbf{v}\cdot\underline{\boldsymbol{\nabla}t_r})\Big],\\
&= \frac{1}{c}\Big[\mathbf{v}+(\boldsymbol{\mathscr{r}}\cdot\mathbf{a}-v^2)\boldsymbol{\nabla}t_r\Big],
\end{aligned} \tag{73}$$

and back to (67), by the results shown above, we can verify:

$$\begin{aligned}
\boldsymbol{\nabla}\mathscr{r} &= \frac{1}{\mathscr{r}}\Big[\boldsymbol{\mathscr{r}}\times\underbrace{(\boldsymbol{\nabla}\times\mathbf{r})}_{=0}-\boldsymbol{\mathscr{r}}\times\underbrace{(\boldsymbol{\nabla}\times\mathbf{w})}_{-\mathbf{v}\times\boldsymbol{\nabla}t_r}+\underbrace{(\boldsymbol{\mathscr{r}}\cdot\boldsymbol{\nabla})\mathbf{r}}_{\boldsymbol{\mathscr{r}}}-\underbrace{(\boldsymbol{\mathscr{r}}\cdot\boldsymbol{\nabla})\mathbf{w}}_{\mathbf{v}(\boldsymbol{\mathscr{r}}\cdot\boldsymbol{\nabla}t_r)}\Big]\\
&= \frac{1}{\mathscr{r}}\Big[\boldsymbol{\mathscr{r}}-\mathbf{v}(\boldsymbol{\mathscr{r}}\cdot\boldsymbol{\nabla}t_r)+\underbrace{\boldsymbol{\mathscr{r}}\times(\mathbf{v}\times\boldsymbol{\nabla}t_r)}_{\text{Bac-Cab Rule}}\Big]\\
&= \frac{1}{\mathscr{r}}\Big[\boldsymbol{\mathscr{r}}-\mathbf{v}(\boldsymbol{\mathscr{r}}\cdot\boldsymbol{\nabla}t_r)+\mathbf{v}(\boldsymbol{\mathscr{r}}\cdot\boldsymbol{\nabla}t_r)-\boldsymbol{\nabla}t_r(\boldsymbol{\mathscr{r}}\cdot\mathbf{v})\Big] = \hat{\boldsymbol{\mathscr{r}}}-\boldsymbol{\nabla}t_r(\hat{\boldsymbol{\mathscr{r}}}\cdot\mathbf{v}).
\end{aligned} \tag{74}$$

Therefore, (66) gives that

$$\boldsymbol{\nabla}t_r = -\frac{1}{c}\boldsymbol{\nabla}\mathscr{r} = \frac{1}{c}\Big[\boldsymbol{\nabla}t_r(\hat{\boldsymbol{\mathscr{r}}}\cdot\mathbf{v})-\hat{\boldsymbol{\mathscr{r}}}\Big] \Rightarrow \boldsymbol{\nabla}t_r = \frac{-\hat{\boldsymbol{\mathscr{r}}}}{1-\hat{\boldsymbol{\mathscr{r}}}\cdot\mathbf{v}/c}. \tag{75}$$

Combining all these shit (75), (73) and (66), we have:

$$\begin{cases}
\boldsymbol{\nabla}\mathscr{r} = -c\boldsymbol{\nabla}t_r = \dfrac{c\hat{\boldsymbol{\mathscr{r}}}}{1-\hat{\boldsymbol{\mathscr{r}}}\cdot\mathbf{v}/c}\\
\dfrac{1}{c}\boldsymbol{\nabla}(\boldsymbol{\mathscr{r}}\cdot\mathbf{v}) = \dfrac{1}{c}\dfrac{1}{(1-\hat{\boldsymbol{\mathscr{r}}}\cdot\mathbf{v}/c)}\Big[(1-\hat{\boldsymbol{\mathscr{r}}}\cdot\mathbf{v}/c)\mathbf{v}-(\boldsymbol{\mathscr{r}}\cdot\mathbf{a}-v^2)\hat{\boldsymbol{\mathscr{r}}}\Big].
\end{cases} \tag{76}$$

Hence,

$$\boldsymbol{\nabla}\Big[\mathscr{r}(1-\hat{\boldsymbol{\mathscr{r}}}\cdot\mathbf{v}/c)\Big] = \frac{1}{c}\frac{\Big[(c^2-v^2+\boldsymbol{\mathscr{r}}\cdot\mathbf{a})\hat{\boldsymbol{\mathscr{r}}}-(1-\hat{\boldsymbol{\mathscr{r}}}\cdot\mathbf{v}/c)\mathbf{v}\Big]}{(1-\hat{\boldsymbol{\mathscr{r}}}\cdot\mathbf{v}/c)}. \tag{77}$$

Substitute this result into (65), and we can obtain the gradient potential:

$$\begin{aligned}
\boldsymbol{\nabla}V &= \frac{1}{4\pi\epsilon_0}\frac{q}{c\mathscr{r}^2(1-\hat{\boldsymbol{\mathscr{r}}}\cdot\mathbf{v}/c)^3}\Big[(1-\hat{\boldsymbol{\mathscr{r}}}\cdot\mathbf{v}/c)\mathbf{v}-(c^2-v^2+\boldsymbol{\mathscr{r}}\cdot\mathbf{a})\hat{\boldsymbol{\mathscr{r}}}\Big]\\
&= \boxed{\frac{1}{4\pi\epsilon_0}\frac{qc}{(\mathscr{r}c-\boldsymbol{\mathscr{r}}\cdot\mathbf{v})^3}\Big[(\mathscr{r}c-\boldsymbol{\mathscr{r}}\cdot\mathbf{v})\mathbf{v}-(c^2-v^2+\boldsymbol{\mathscr{r}}\cdot\mathbf{a})\boldsymbol{\mathscr{r}}\Big]}
\end{aligned} \tag{78}$$

Part 2. The second term $\dfrac{\partial \mathbf{A}}{\partial t}$:
We've known that

$$\mathbf{A} = \frac{1}{c}\frac{1}{4\pi\epsilon_0}\frac{q\mathbf{v}}{(\mathscr{r}c - \boldsymbol{\mathscr{r}}\cdot\mathbf{v})}, \tag{79}$$

and by (42), the differentiation of $\mathbf{A}$ with respect to time t is

$$\frac{\partial \mathbf{A}}{\partial t} = \frac{\partial \mathbf{A}}{\partial t_r}\frac{\partial t_r}{\partial t} = \frac{1}{(1-\hat{\boldsymbol{\mathscr{r}}}\cdot\mathbf{v}/c)}\frac{\partial \mathbf{A}}{\partial t_r} = \frac{\mathscr{r}c}{(\mathscr{r}c - \boldsymbol{\mathscr{r}}\cdot\mathbf{v})}\frac{\partial \mathbf{A}}{\partial t_r}. \tag{80}$$

Then we can start to differentiate it:

$$\begin{aligned}
\frac{\partial \mathbf{A}}{\partial t_r} &= \frac{1}{4\pi\epsilon_0 c}\frac{q\dfrac{\mathrm{d}\mathbf{v}}{\mathrm{d}t_r}(\mathscr{r}c - \boldsymbol{\mathscr{r}}\cdot\mathbf{v}) - q\mathbf{v}\left[c\dfrac{\mathrm{d}\mathscr{r}}{\mathrm{d}t_r} - \dfrac{\mathrm{d}}{\mathrm{d}t_r}(\boldsymbol{\mathscr{r}}\cdot\mathbf{v})\right]}{(\mathscr{r}c - \boldsymbol{\mathscr{r}}\cdot\mathbf{v})^2} \\
&= \frac{1}{4\pi\epsilon_0 c}\frac{q\mathbf{a}(\mathscr{r}c - \boldsymbol{\mathscr{r}}\cdot\mathbf{v}) - q\mathbf{v}\left[-c\mathbf{v}\cdot\boldsymbol{\mathscr{r}}/\mathscr{r} - \left(\dfrac{\mathrm{d}\boldsymbol{\mathscr{r}}}{\mathrm{d}t_r}\cdot\mathbf{v} + \boldsymbol{\mathscr{r}}\cdot\dfrac{\mathrm{d}\mathbf{v}}{\mathrm{d}t_r}\right)\right]}{(\mathscr{r}c - \boldsymbol{\mathscr{r}}\cdot\mathbf{v})^2} \\
&= \frac{1}{4\pi\epsilon_0 c}\frac{q\mathbf{a}(\mathscr{r}c - \boldsymbol{\mathscr{r}}\cdot\mathbf{v}) - q\mathbf{v}\left[-c\mathbf{v}\cdot\boldsymbol{\mathscr{r}}/\mathscr{r} - \left(-v^2 + \boldsymbol{\mathscr{r}}\cdot\mathbf{a}\right)\right]}{(\mathscr{r}c - \boldsymbol{\mathscr{r}}\cdot\mathbf{v})^2} \\
&= \frac{qc}{4\pi\epsilon_0 c}\frac{(\mathscr{r} - \boldsymbol{\mathscr{r}}\cdot\mathbf{v}/c)\mathbf{a} + \left[\mathbf{v}\cdot\boldsymbol{\mathscr{r}}/\mathscr{r} - v^2/c + \boldsymbol{\mathscr{r}}\cdot\mathbf{a}/c\right]\mathbf{v}}{(\mathscr{r}c - \boldsymbol{\mathscr{r}}\cdot\mathbf{v})^2},
\end{aligned}$$

and then substitute the result into (80), we obtain:

$$\begin{aligned}
\frac{\partial \mathbf{A}}{\partial t} &= \frac{\mathscr{r}c}{(\mathscr{r}c - \boldsymbol{\mathscr{r}}\cdot\mathbf{v})}\frac{q}{4\pi\epsilon_0}\frac{(\mathscr{r} - \boldsymbol{\mathscr{r}}\cdot\mathbf{v}/c)\mathbf{a} + \left[\mathbf{v}\cdot\boldsymbol{\mathscr{r}}/\mathscr{r} - v^2/c + \boldsymbol{\mathscr{r}}\cdot\mathbf{a}/c\right]\mathbf{v}}{(\mathscr{r}c - \boldsymbol{\mathscr{r}}\cdot\mathbf{v})^2} \\
&= \frac{1}{4\pi\epsilon_0}\frac{qc}{(\mathscr{r}c - \boldsymbol{\mathscr{r}}\cdot\mathbf{v})^3}\left[\mathscr{r}^2\mathbf{a} - \frac{\mathscr{r}}{c}(\boldsymbol{\mathscr{r}}\cdot\mathbf{v})\mathbf{a} + (\boldsymbol{\mathscr{r}}\cdot\mathbf{v})\mathbf{v} - \frac{\mathscr{r}}{c}v^2\mathbf{v} + \frac{\mathscr{r}}{c}(\boldsymbol{\mathscr{r}}\cdot\mathbf{a})\mathbf{v}\right].
\end{aligned} \tag{81}$$

Ugly! But we can compare this with $\boldsymbol{\nabla}V$, and think that we may build the terms that are similar to $\boldsymbol{\nabla}V$. Namely, the expression may contain $(\mathscr{r}c - \boldsymbol{\mathscr{r}}\cdot\mathbf{v})$ and

$(c^2 - v^2 + \boldsymbol{\mathscr{r}} \cdot \mathbf{a})$. So, just check the term:

$$\mathscr{r}^2\mathbf{a} - \frac{\mathscr{r}}{c}(\boldsymbol{\mathscr{r}} \cdot \mathbf{v})\mathbf{a} + (\boldsymbol{\mathscr{r}} \cdot \mathbf{v})\mathbf{v} - \frac{\mathscr{r}}{c}v^2\mathbf{v} + \frac{\mathscr{r}}{c}(\boldsymbol{\mathscr{r}} \cdot \mathbf{a})\mathbf{v}$$

$$= \mathscr{r}^2\mathbf{a} + \big(-\boldsymbol{\mathscr{r}} \cdot \mathbf{v}\big)\big(\mathscr{r}\mathbf{a}/c - \mathbf{v}\big) + \frac{\mathscr{r}}{c}\big(-v^2 + \boldsymbol{\mathscr{r}} \cdot \mathbf{a}\big)\mathbf{v}$$

$$= \mathscr{r}^2\mathbf{a} + \big(-\boldsymbol{\mathscr{r}} \cdot \mathbf{v}\big)\big(\mathscr{r}\mathbf{a}/c - \mathbf{v}\big) + \frac{\mathscr{r}}{c}\big(c^2 - v^2 + \boldsymbol{\mathscr{r}} \cdot \mathbf{a}\big)\mathbf{v} - \frac{\mathscr{r}}{c} \times c^2\mathbf{v}$$

$$= \mathscr{r}c \cdot \frac{\mathscr{r}}{c}\mathbf{a} - \mathscr{r}c\mathbf{v} + \big(-\boldsymbol{\mathscr{r}} \cdot \mathbf{v}\big)\big(\mathscr{r}\mathbf{a}/c - \mathbf{v}\big) + \frac{\mathscr{r}}{c}\big(c^2 - v^2 + \boldsymbol{\mathscr{r}} \cdot \mathbf{a}\big)\mathbf{v}$$

$$= \big(\mathscr{r}c\big)\big(\mathscr{r}\mathbf{a}/c - \mathbf{v}\big) + \big(-\boldsymbol{\mathscr{r}} \cdot \mathbf{v}\big)\big(\mathscr{r}\mathbf{a}/c - \mathbf{v}\big) + \frac{\mathscr{r}}{c}\big(c^2 - v^2 + \boldsymbol{\mathscr{r}} \cdot \mathbf{a}\big)\mathbf{v}$$

$$= \boxed{\big(\mathscr{r}c - \boldsymbol{\mathscr{r}} \cdot \mathbf{v}\big)\big(\mathscr{r}\mathbf{a}/c - \mathbf{v}\big) + \frac{\mathscr{r}}{c}\big(c^2 - v^2 + \boldsymbol{\mathscr{r}} \cdot \mathbf{a}\big)\mathbf{v}}.$$

As a result, substitute this into the original expression (81) and obtain:

$$\boxed{\begin{aligned}\frac{\partial \mathbf{A}}{\partial t} = \frac{1}{4\pi\epsilon_0}\frac{qc}{(\mathscr{r}c - \boldsymbol{\mathscr{r}} \cdot \mathbf{v})^3}\Big[&\big(\mathscr{r}c - \boldsymbol{\mathscr{r}} \cdot \mathbf{v}\big)\big(\mathscr{r}\mathbf{a}/c - \mathbf{v}\big) \\ &+ \frac{\mathscr{r}}{c}\big(c^2 - v^2 + \boldsymbol{\mathscr{r}} \cdot \mathbf{a}\big)\mathbf{v}\Big],\end{aligned}} \tag{82}$$

where we define $\mathbf{u} = c\hat{\boldsymbol{\mathscr{r}}} - \mathbf{v}$, so

$$\mathscr{r}c - \boldsymbol{\mathscr{r}} \cdot \mathbf{v} = c\hat{\boldsymbol{\mathscr{r}}} \cdot \boldsymbol{\mathscr{r}} - \mathbf{v} \cdot \boldsymbol{\mathscr{r}} = \big(c\hat{\boldsymbol{\mathscr{r}}} - \mathbf{v}\big) \cdot \boldsymbol{\mathscr{r}} = \mathbf{u} \cdot \boldsymbol{\mathscr{r}}. \tag{83}$$

Finally, combining (82) with (78), the electric fields of a moving charge is therefore:

$$\mathbf{E}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0}\frac{qc}{(\mathscr{r}c - \boldsymbol{\mathscr{r}} \cdot \mathbf{v})^3}\Big[(c^2 - v^2 + \boldsymbol{\mathscr{r}} \cdot \mathbf{a})\boldsymbol{\mathscr{r}} - (\mathscr{r}c - \boldsymbol{\mathscr{r}} \cdot \mathbf{v})\mathbf{v}$$

$$- \big(\mathscr{r}c - \boldsymbol{\mathscr{r}} \cdot \mathbf{v}\big)\big(\mathscr{r}\mathbf{a}/c - \mathbf{v}\big) - \frac{\mathscr{r}}{c}\big(c^2 - v^2 + \boldsymbol{\mathscr{r}} \cdot \mathbf{a}\big)\mathbf{v}\Big]$$

$$= \frac{1}{4\pi\epsilon_0}\frac{qc}{(\mathbf{u} \cdot \boldsymbol{\mathscr{r}})^3}\Big[\big(c^2 - v^2 + \boldsymbol{\mathscr{r}} \cdot \mathbf{a}\big)\Big(\underbrace{\boldsymbol{\mathscr{r}}}_{\mathscr{r}\hat{\boldsymbol{\mathscr{r}}}} - \frac{\mathscr{r}}{c}\mathbf{v}\Big) - (\mathbf{u} \cdot \boldsymbol{\mathscr{r}})\big(\mathbf{v} + \mathscr{r}\mathbf{a}/c - \mathbf{v}\big)\Big]$$

$$= \frac{1}{4\pi\epsilon_0}\frac{qc}{(\mathbf{u} \cdot \boldsymbol{\mathscr{r}})^3}\frac{\mathscr{r}}{c}\Big[(c^2 - v^2 + \boldsymbol{\mathscr{r}} \cdot \mathbf{a})(c\hat{\boldsymbol{\mathscr{r}}} - \mathbf{v}) - (\mathbf{u} \cdot \boldsymbol{\mathscr{r}})\mathbf{a}\Big]$$

$$= \frac{1}{4\pi\epsilon_0}\frac{q\mathscr{r}}{(\mathbf{u} \cdot \boldsymbol{\mathscr{r}})^3}\Big[(c^2 - v^2)\mathbf{u} + \mathbf{u}(\boldsymbol{\mathscr{r}} \cdot \mathbf{a}) - \mathbf{a}(\mathbf{u} \cdot \boldsymbol{\mathscr{r}})\Big] \quad \text{B(AC)-C(AB) Rule}$$

After these series of ugly derivation, we can finally determine the electric field of a moving charge with all its information. I,e,

$$\boxed{\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\frac{q\boldsymbol{\mathscr{r}}}{(\mathbf{u}\cdot\boldsymbol{\mathscr{r}})^3}\Big[\underbrace{(c^2-v^2)\mathbf{u}}_{\text{Distortion}} - \underbrace{\boldsymbol{\mathscr{r}}\times(\mathbf{u}\times\mathbf{a})}_{\text{Radiation}}\Big].} \tag{84}$$

(1) $\underline{(c^2-v^2)\mathbf{u}}$: Geometrical Distortion.

This term represents the velocity field (or generalized Coulomb field). It originates purely from the motion (velocity) of the charge, even if the charge is not accelerating.

- Its spatial dependence behaves as $\sim \frac{1}{R^2}$, hence it dominates in the near-field region.
- Physically, it can be interpreted as the Coulomb field distorted by relativistic motion, where the field lines are no longer spherically symmetric.
- This term does not transport energy to infinity, meaning it does not correspond to radiation.

(2) $\underline{\boldsymbol{\mathscr{r}}\times(\mathbf{u}\times\mathbf{a})}$: Radiation

This term represents the radiation field, which arises from the acceleration of the charge.

- Its spatial dependence behaves as $\sim \frac{1}{R}$, which is the characteristic scaling required for energy propagation.
- The structure $\boldsymbol{\mathscr{r}}\times(\mathbf{u}\times\mathbf{a})$ extracts the transverse component of acceleration, ensuring the field is perpendicular to the propagation direction.
- This term is responsible for electromagnetic radiation, i.e., it carries energy outward via the Poynting vector.

Meanwhile, the corresponding magnetic field is that

$$\mathbf{B}(\mathbf{r},t) = \boldsymbol{\nabla}\times\mathbf{A} = \boldsymbol{\nabla}\times\left(\frac{\mathbf{v}}{c^2}V\right) = \frac{1}{c^2}\Big[V(\boldsymbol{\nabla}\times\mathbf{v}) - \mathbf{v}\times(\boldsymbol{\nabla}V)\Big], \tag{85}$$

and we substitute (69) and (78) into the curl of $\mathbf{v}$ and the gradient of V, we have

$$\begin{aligned}\frac{1}{c^2}\Big[V(\boldsymbol{\nabla}\times\mathbf{v}) - \mathbf{v}\times(\boldsymbol{\nabla}V)\Big] = \frac{1}{c^2}\Bigg[&\frac{1}{4\pi\epsilon_0}\frac{qc}{(\mathbf{u}\cdot\boldsymbol{\mathscr{r}})}(-\mathbf{a}\times\boldsymbol{\nabla}t_r) \qquad \boldsymbol{\nabla}t_r = \frac{-\boldsymbol{\mathscr{r}}}{\mathbf{u}\cdot\boldsymbol{\mathscr{r}}}\\ &-\mathbf{v}\times\left(\frac{1}{4\pi\epsilon_0}\frac{qc}{(\mathbf{u}\cdot\boldsymbol{\mathscr{r}})^3}\Big[(\mathbf{u}\cdot\boldsymbol{\mathscr{r}})\mathbf{v} - (c^2-v^2+\boldsymbol{\mathscr{r}}\cdot\mathbf{a})\boldsymbol{\mathscr{r}}\Big]\right)\Bigg]\end{aligned}$$

By factoring out the mutual terms, then we can keep evaluating:

$$\begin{aligned}\cdots &= \frac{1}{c^2}\frac{qc}{4\pi\epsilon_0}\frac{1}{(\mathbf{u}\cdot\boldsymbol{\mathscr{r}})^3}\Big[(\mathbf{u}\cdot\boldsymbol{\mathscr{r}})\mathbf{a}\times\boldsymbol{\mathscr{r}} + (c^2 - v^2 + \boldsymbol{\mathscr{r}}\cdot\mathbf{a})\mathbf{v}\times\boldsymbol{\mathscr{r}}\Big] \\ &= -\frac{1}{c}\frac{q}{4\pi\epsilon_0}\frac{1}{(\mathbf{u}\cdot\boldsymbol{\mathscr{r}})^3}\underbrace{\boldsymbol{\mathscr{r}}}_{\mathscr{r}\hat{\boldsymbol{\mathscr{r}}}}\times\Big[(c^2-v^2)\mathbf{v} + (\boldsymbol{\mathscr{r}}\cdot\mathbf{a})\mathbf{v} + (\mathbf{u}\cdot\boldsymbol{\mathscr{r}})\mathbf{a}\Big],\end{aligned}$$

since $\mathbf{u} = c\hat{\boldsymbol{\mathscr{r}}} - \mathbf{v}$ and it's crossed into $\boldsymbol{\mathscr{r}}$, it turns out that $\boldsymbol{\mathscr{r}}\times c\hat{\boldsymbol{\mathscr{r}}} = 0$. Hence, it's equivalent that

$$\boldsymbol{\mathscr{r}}\times\mathbf{u} = \boldsymbol{\mathscr{r}}\times(c\hat{\boldsymbol{\mathscr{r}}} - \mathbf{v}) = -\boldsymbol{\mathscr{r}}\times\mathbf{v} \Rightarrow \mathbf{v}\mapsto -\mathbf{u}. \tag{86}$$

Then it becomes

$$\begin{aligned}&-\frac{1}{c}\frac{q}{4\pi\epsilon_0}\frac{1}{(\mathbf{u}\cdot\boldsymbol{\mathscr{r}})^3}\underbrace{\boldsymbol{\mathscr{r}}}_{\mathscr{r}\hat{\boldsymbol{\mathscr{r}}}}\times\Big[(c^2-v^2)\mathbf{v} + (\boldsymbol{\mathscr{r}}\cdot\mathbf{a})\mathbf{v} + (\mathbf{u}\cdot\boldsymbol{\mathscr{r}})\mathbf{a}\Big] \\ &= \frac{1}{c}\hat{\boldsymbol{\mathscr{r}}}\times\frac{1}{4\pi\epsilon_0}\frac{q\mathscr{r}}{(\mathbf{u}\cdot\boldsymbol{\mathscr{r}})^3}\Big[(c^2-v^2)\mathbf{u} + (\boldsymbol{\mathscr{r}}\cdot\mathbf{a})\mathbf{u} - (\mathbf{u}\cdot\boldsymbol{\mathscr{r}})\mathbf{a}\Big] \\ &= \frac{1}{c}\hat{\boldsymbol{\mathscr{r}}}\times\underbrace{\left[\frac{1}{4\pi\epsilon_0}\frac{q\hat{\boldsymbol{\mathscr{r}}}}{(\mathbf{u}\cdot\boldsymbol{\mathscr{r}})^3}\Big[(c^2-v^2)\mathbf{u} + \boldsymbol{\mathscr{r}}\times(\mathbf{u}\times\mathbf{a})\Big]\right]}_{\mathbf{E}(\mathbf{r},t)},\end{aligned} \tag{87}$$

hence, the relation between the electric field and the magnetic field built from the moving charge is:

$$\boxed{\mathbf{B}(\mathbf{r},t) = \frac{1}{c}\hat{\boldsymbol{\mathscr{r}}}\times\mathbf{E}(\mathbf{r},t).} \tag{88}$$